

Gabriel Alexander Karp-Gershon

Product & Project Expert • Electro-Optics • System Architecture

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TOOLS & METHODS

Project workflow: Jira, Confluence, Codebeamer ALM

Architecture: System architecture, requirements & traceability, MBSE

Methods: Lean, Six Sigma Black Belt, FMEA, DOE, RFQ/RFI evaluation

Reporting & data: Power BI, Excel, SQL, Python

Engineering: Python, MATLAB, LabVIEW, Docker

Domain: Electro-optics, machine vision, LiDAR, photonics, SPAD, SiPM, TCSPC, optical metrology, semiconductor physics, single-photon detection

REGULATORY CONTEXT

Systems, safety and cybersecurity

ASPICE: Requirements, traceability

PROFILE

Hardware project lead with 15 years in automotive LiDAR, consumer cameras, and defence electro-optics. Built modular hardware programmes from requirements to production ramp, including supplier qualification, verification, and certification. Experience spans optics, illumination, detection, FPGA/SoC, and software interfaces under Automotive SPICE and ISO 26262.

PROFESSIONAL EXPERIENCE

Founder & Product / Project Expert | Kanzen

- Founded a consultancy bridging hardware product development and technical-commercial evaluation; clients in deep-tech.
- Led RFQ and RFI evaluation programmes: structured supplier scoring on quality, lead time, traceability, and total landed cost.
- Translated measured system behaviour into product decisions, delivering executable engineering scope for stakeholders.

System Architect & Change Control Lead | Innoviz

- Owned change governance for the LiDAR product line under ASPICE and ISO 26262 traceability requirements.
- Coordinated cross-team change requests from OEM customers across optics, electronics, firmware, and validation.
- Authored architectural decision records and validation criteria that fed directly into V-model verification plans.

REGULATORY CONTEXT (Cont.)

Electrical and EMC

DIN EN 61010: Electrical safety, lab & measurement

CISPR 25: EMC emissions

ISO 11452: EMC immunity

Environmental, durability and materials compliance

MIL-STD-810G: Environmental test methods

IEC 60529: Ingress protection (IP)

PROFESSIONAL EXPERIENCE (Cont.)

Customer Change Requests Specialist | Innoviz

- Single point of contact for OEM-driven change requests; converted commercial expectations into testable engineering scope.
- Ran weekly cross-functional reviews; tracked KPIs on cycle time, rework, and customer-found defects.
- Reduced rework on late-stage changes by formalising the gate criteria for system-level acceptance.

System Architect | Innoviz

- Defined the system architecture for automotive LiDAR: optics, electronics, FPGA/SoC, and software interfaces.
- Specified component-level requirements and validation methods; aligned them with ASPICE and ISO 26262 expectations.
- Led design reviews and architectural trade studies for next-generation product variants.

AI-assisted — author retains responsibility for content.

ADDITIONAL INFORMATION

Languages: English (full professional), Hebrew (native), Spanish (professional), Danish (B1)

ADDITIONAL INFORMATION (CONT.)

Volunteer: Operations Manager & Assistant Coach, Copenhagen Wolves RFC (Pan Idræt Rugby)

ADDITIONAL INFORMATION (CONT.)

Hobbies: Rugby, Hiking, tai-chi, reading, supervision of three feline napping experts